IP Networks for IoT: IPv4, NAT, IPv6 and Neighbor Discovery

Communication Technologies for the Internet of Things, Lecture 08

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Agenda (1/2)

- The network layer and IPv4 addressing
- Address space exhaustion and its causes
- Mitigation: classless addressing and NAT
- The limitations of NAT
- IPv6: features, addressing and address types
- Autoconfiguration and interface identifiers

Agenda (2/2)

- Neighbor Discovery Protocol and ICMPv6
- Routing, DHCPv6 and DNS in IPv6

The Network Layer (1/2)

- The network layer is responsible for forwarding information across the systems it has to traverse until it reaches its destination.
- It also takes on error control and flow control between the two ends of the communication.

The Network Layer (2/2)

- This is where the IP protocol lives, the element that gives the Internet its unity: very different link technologies interoperate because they share the same network protocol.
- In an IoT solution, this layer is the point where data collected by short-range technologies stops being confined to the local network and can reach any destination.

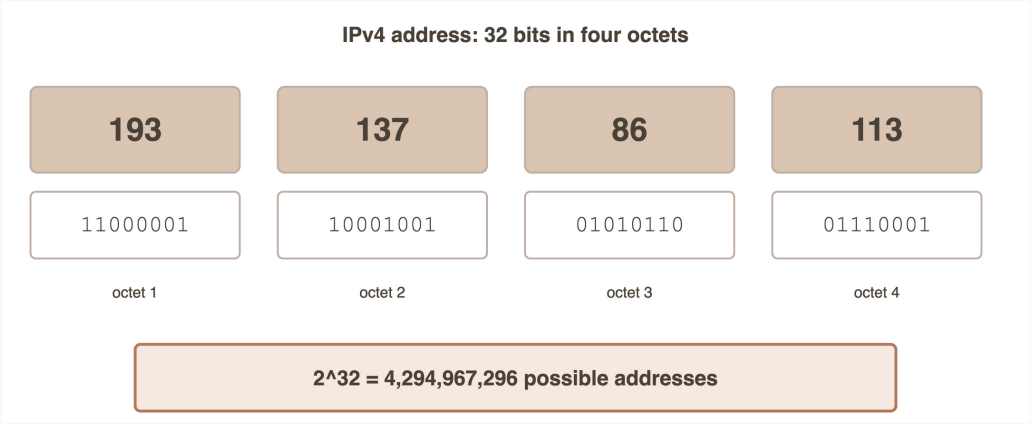
IPv4 Addressing (1/2)

- Every machine on the network must have an address that identifies it uniquely.
- In version 4 of the protocol, addresses are 32 bits long.
- The usual notation is decimal, with the four octets separated by dots: the address 193.137.86.113 corresponds to the binary sequence 11000001 10001001 01010110 01110001.

IPv4 Addressing (2/2)

- A 32-bit space can represent 2^{32}, that is, about 4,294,967,296 distinct addresses.
- This number, which seemed more than enough when the protocol was defined, proved insufficient at the scale of today's Internet.

The IPv4 Address in Binary



Causes of Address Exhaustion (1/2)

- **Mobile devices:** mobile phones became full computers, and 4G specifications require IP addressing.
- **Always-on connections:** home equipment stopped being switched off, keeping addresses occupied continuously; on mobile devices, only battery life imposes some interruption.
- **Internet demographics:** hundreds of millions of homes came online, and developing countries joined the network progressively.

Causes of Address Exhaustion (2/2)

- **Internet of Things:** connecting sensors and actuators multiplies, by several orders of magnitude, the number of devices that need an address.
- This last cause is the one that makes the problem directly relevant to this course.

Mitigation Strategies (1/2)

- The predicted exhaustion of the address space did not paralyse the Internet, because stop-gap solutions were developed.
- **Classless addressing:** allows address blocks sized to actual need, instead of the original rigid classes, reducing waste.
- **NAT (Network Address Translation):** lets several machines on a private network share a single public address.

Mitigation Strategies (2/2)

- In parallel, a structural solution was developed: IPv6 addressing, with enough space never to raise the problem again.
- The distinction between patch and solution matters: the first two postpone the problem, the third resolves it.

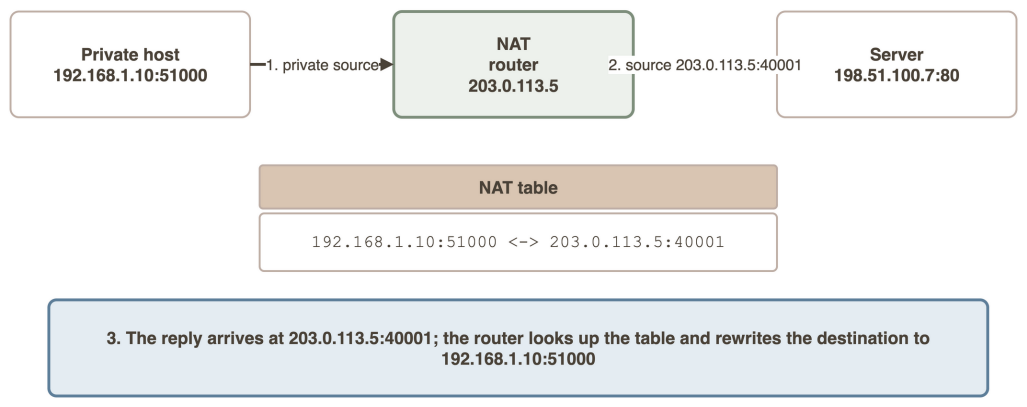
NAT: How It Works (1/2)

- When a packet reaches the router from the private network, the router replaces the source address in the IP header with its own outgoing public address.
- Source and destination ports may also be changed in the process.
- The translation is recorded in the router's NAT table, where all connections between the public and private networks are noted.
- When connections are closed, the corresponding entry is removed from the table.

NAT: How It Works (2/2)

- When the reply arrives, the router consults the table and performs the inverse operation: the destination address is replaced by the corresponding private address.
- The concrete behaviour of the table depends on the NAT mode configured on the device.

Address Translation in Practice



Limitations of NAT (1/2)

- It breaks the one-machine-one-address principle on which the original design of the Internet was based.
- The simplest implementations do not support protocols such as FTP or H.323; implementations that do support them have to read information from higher layers.
- It does not support traffic other than TCP or UDP; some implementations work around the limitation, at the cost of performance.
- It violates the principle that a layer only talks to adjacent layers, by requiring the network layer to interpret application content.

Limitations of NAT (2/2)

- It increases latency, by adding processing to every packet.
- Paradoxically, the success of NAT in postponing the problem is one of the reasons why IPv6 took so long to be adopted.

Port NAT and the Server Problem (1/2)

- Conventional NAT lets internal machines initiate connections to the outside, but not external machines reach an internal service.
- Port NAT partially solves the problem by forwarding a port of a public address to a port of a private address.
- The configuration is done port by port, explicitly associating each internal service with an external port.

Port NAT and the Server Problem (2/2)

- The limitation is obvious: the solution serves one device at a time, and does not scale to an installation with several sensors or actuators reachable from outside.
- This is a concrete problem in IoT, where it is common to need to reach individual devices from the network.

IPv6 Adoption (1/2)

- The IETF IPv6 working group was set up in the early 1990s precisely to solve the addressing problem.
- Actual adoption, however, was much slower than expected, for three main reasons.
- The development of classless routing (CIDR) and of NAT reduced the urgency of the problem, easing the pressure that would motivate migration.
- The lack of technical knowledge about the new protocol was a barrier in many organisations.

IPv6 Adoption (2/2)

- The cost of replacing equipment that did not support the protocol weighed on investment decisions.
- Adoption remains uneven between countries, and this unevenness has practical consequences when designing solutions for different markets.

IPv6 Features (1/2)

- The enlarged address space allows global addressing, flexibility in allocation, route aggregation, *multi-homing*, autoconfiguration and network renumbering.
- The header was simplified relative to IPv4, which translates into more efficient forwarding, better performance and better scalability.
- While IPv4 uses 4 bytes, or 32 bits, with about 4.29 billion possible addresses, IPv6 uses 16 bytes, or 128 bits.

IPv6 Features (2/2)

- The resulting space is 340,282,366,920,938,463,463,374,607,431,768,211,456 addresses, a number that removes addressing from the list of design constraints.
- For the Internet of Things, this abundance means that every sensor can have its own globally routable address, removing the need for address translation.

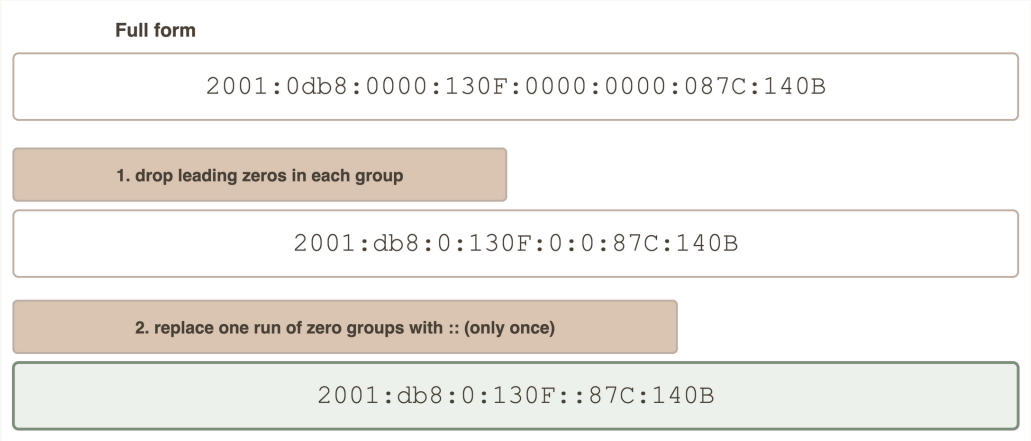
Representing IPv6 Addresses (1/2)

- Addresses are written as groups of 16-bit hexadecimal numbers.
- Hexadecimal numbers are not case sensitive.
- Groups are separated by colons.
- Abbreviations are allowed: leading zeros within a group may be omitted.

Representing IPv6 Addresses (2/2)

- A contiguous sequence of zero groups may be replaced by two consecutive colons, an abbreviation that can be used only once per address, to avoid ambiguity.
- Thus the address 2001:0db8:0000:130F:0000:0000:087C:140B can be written in abbreviated form as 2001:db8:0:130F::87C:140B.

Abbreviating an IPv6 Address



IPv6 Address Types (1/2)

- **Unicast:** address of a single interface, with one-to-one delivery to that interface.
- **Multicast:** address of a set of interfaces, with one-to-many delivery to all interfaces in the set.
- **Anycast:** address of a set of interfaces, with delivery to a single interface of the set, the nearest one according to routing criteria.

IPv6 Address Types (2/2)

- The broadcast address, which exists in IPv4, no longer exists in IPv6.
- Its function was absorbed by multicast, with the advantage that traffic is processed only by the nodes actually interested, instead of interrupting every device on the network.

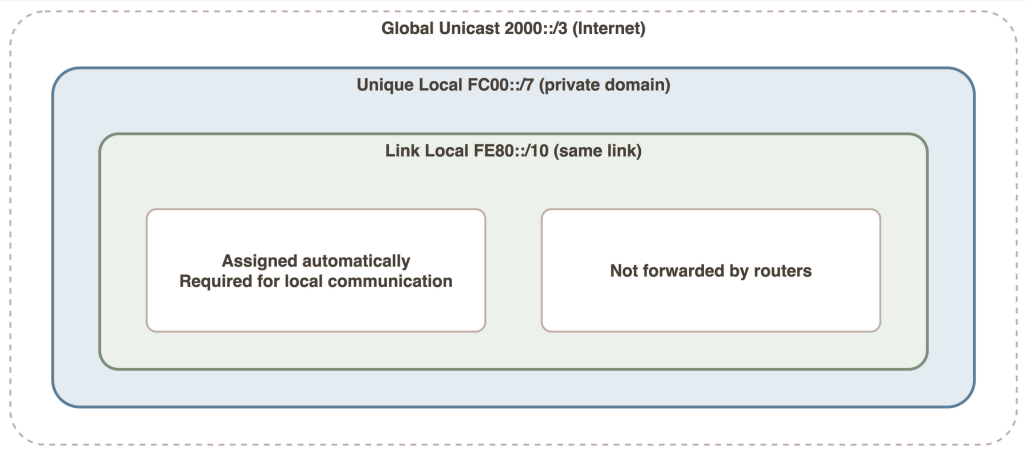
Addressing Model and Scope (1/2)

- In IPv6, each interface typically has several addresses at the same time, not just one.
- Addresses have an associated *scope*, which delimits the region where they are valid.
- **Link Local:** valid only on the same local network or link. They are assigned automatically as soon as IPv6 is enabled and are required for local communication between two devices.

Addressing Model and Scope (2/2)

- **Unique Local:** valid within the same private domain, routable only inside the same autonomous system, and not usable on the Internet. They are used for local communications and for VPNs between sites.
- **Global Unicast:** globally routable addresses, with a hierarchical structure that facilitates route aggregation.
- Addresses also have an associated lifetime, with default values, which allows orderly renumbering of a network.

IPv6 Address Scopes



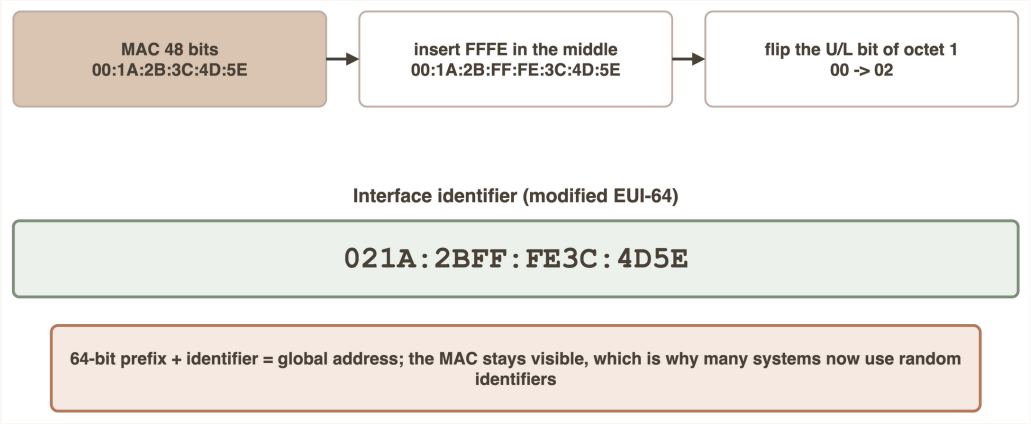
Interface Identifier and EUI-64 Format (1/2)

- The 64 low-order bits of any IPv6 address are the interface identifier, that is, the second half of the address.
- This identifier can be autoconfigured from a 64-bit EUI-64 identifier, or expanded from a 48-bit MAC address.
- It can also be an automatically generated pseudo-random number, be assigned by DHCP, or be configured manually.
- In stateless autoconfiguration from the MAC address, the 48-bit address is expanded to 64 bits by inserting the value FFFE in the middle 16 bits.

Interface Identifier and EUI-64 Format (2/2)

- The procedure guarantees uniqueness, since it starts from a unique MAC address, and the “u” bit is set to 1 for global scope and to 0 for local scope.
- Deriving the identifier from the MAC address raises privacy concerns, since it makes the device traceable across networks, which is why many systems now use random identifiers.

From a MAC Address to an EUI-64 Identifier



Anycast and Multicast Addresses (1/2)

- An anycast address is assigned to a set of interfaces, which typically belong to different nodes.
- A packet sent to an anycast address is delivered to the nearest interface, determined by routing and propagation times.
- Anycast addresses can only be used by routers, not by end hosts, and cannot be used as the source address of a packet.
- The nodes to which an anycast address is assigned must be explicitly configured to recognise it.

Anycast and Multicast Addresses (2/2)

- Multicast addresses have the prefix FF00::/8, and the second octet defines the lifetime and the scope of the address.
- Among the most used link-scope multicast addresses are FF02::1, which designates all nodes, and FF02::2, which designates all routers.

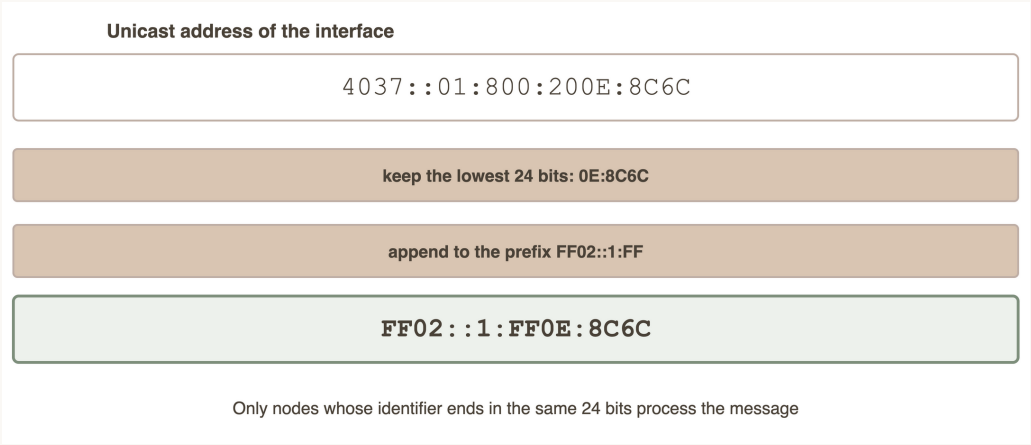
Solicited-Node Multicast Address (1/2)

- Each unicast or anycast address configured on an interface has a corresponding solicited-node multicast address.
- This address is formed by the prefix FF02::1:FF followed by the 24 low-order bits of the interface identifier.
- It has meaning only within the local link scope.
- It is used in Neighbor Solicitation messages, both for link-layer address resolution and for duplicate address detection.

Solicited-Node Multicast Address (2/2)

- The advantage of this mechanism is efficiency: instead of interrupting every node on the network, as a broadcast would, the solicitation is processed only by nodes whose identifier ends in the same 24 bits.
- Random or assigned interface identifiers can result in global and link-local addresses that are equivalent for the purposes of this mechanism.

Building the Solicited-Node Address



ICMPv6 (1/2)

- The Internet Control Message Protocol version 6 is the implementation of ICMP for IPv6, defined in RFC 4443.
- Unlike in IPv4, ICMPv6 is an integral part of the protocol, not an optional add-on: disabling it prevents the network from working normally.
- It keeps the functions of the original ICMP and adds several others.
- It replaces and improves ARP, through the implementation of the Neighbor Discovery Protocol.

ICMPv6 (2/2)

- It is used by hosts to discover routers and to perform address autoconfiguration.
- It also supports duplicate address detection and the management of multicast groups.

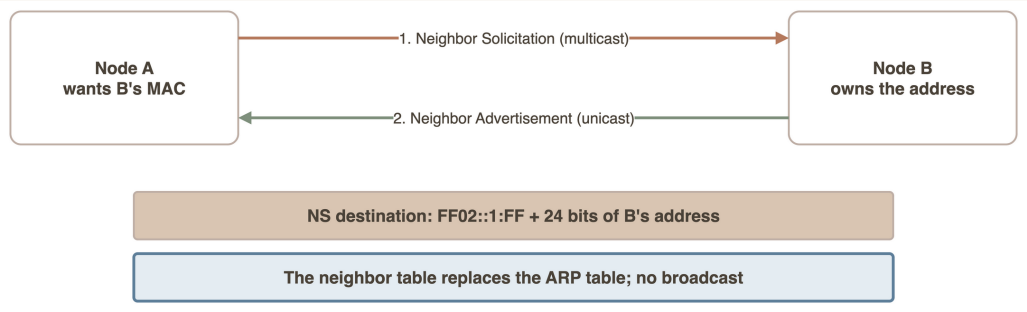
Link-Layer Address Resolution: from ARP to NDP (1/2)

- In IPv6, the ARP protocol no longer exists.
- The table formerly called the ARP table is now called the NDP table, managed by the Neighbor Discovery Protocol.
- NDP keeps the list of known neighbours, associating IPv6 addresses with their MAC addresses.
- It uses ICMPv6 Neighbor Solicitation and Neighbor Advertisement messages.

Link-Layer Address Resolution: from ARP to NDP (2/2)

- To resolve an address, a Neighbor Solicitation message is sent to the solicited-node multicast address of the destination machine.
- The target machine replies with a Neighbor Advertisement containing its link-layer address.

Address Resolution with NDP



Neighbor Discovery Messages (1/2)

- Neighbor Discovery messages are ICMPv6 messages, originated by nodes within the local link scope, with the hop limit set to 255, a mechanism that guarantees they cannot originate outside the local link.
- Each message is made up of an IPv6 header, an ICMPv6 header, a Neighbor Discovery header and its options.
- **Router Solicitation (type 133):** sent by a host to find out whether routers are present on the link.

Neighbor Discovery Messages (2/2)

- **Router Advertisement (type 134):** sent periodically by routers, or in response to a solicitation, including autoconfiguration information and a preference level.
- **Neighbor Solicitation (type 135):** sent to discover the link-layer address of an IPv6 node.
- **Neighbor Advertisement (type 136):** reply to a solicitation, also used to announce changes of the link-layer address.
- **Redirect:** used by a router to signal that a better path exists to a given destination.

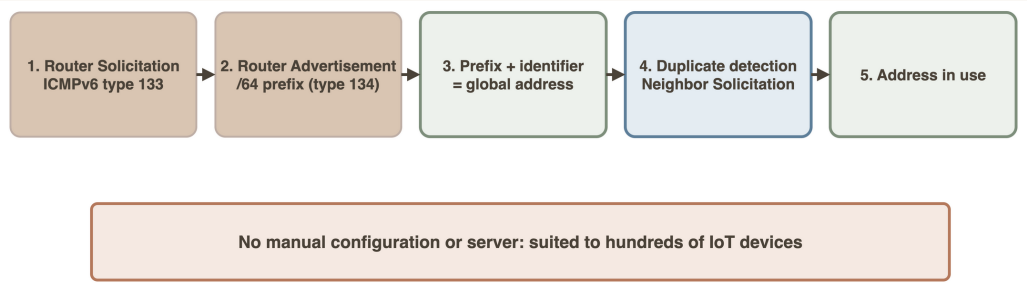
Address Autoconfiguration (1/2)

- **Stateless autoconfiguration:** a node on the link can automatically configure global IPv6 addresses, by appending its 64-bit interface identifier to the 64-bit prefixes received in Router Advertisement messages.
- Additional network information can be obtained through extra fields of those messages, or by using a stateless DHCPv6 server.
- **Stateful autoconfiguration:** addresses are obtained from a DHCPv6 server, which keeps a record of the assignments made.

Address Autoconfiguration (2/2)

- Before using an autoconfigured address, the node runs duplicate address detection, sending a Neighbor Solicitation for the address it intends to use.
- This mechanism is particularly relevant in IoT: it lets hundreds of devices obtain a valid address without any manual configuration or dedicated server.

Stateless Autoconfiguration



Multicast Listener Discovery

- The MLD protocol allows multicast groups to be created and managed in IPv6.
- It is used by an IPv6 router to discover the presence of multicast listeners on directly attached links.
- It also allows determining which multicast addresses those neighbouring nodes are interested in.
- Nodes use MLD to declare interest in specific multicast addresses, including the solicited-node multicast addresses that concern them.
- Without this mechanism, network switches would have to forward all multicast traffic out of every port, cancelling the advantage of multicast over broadcast.

Routing in IPv6 (1/2)

- **Static routing:** routes are defined manually, and may specify only the outgoing interface, only the next hop, or both.
- Floating static routes serve as backup routes, with lower priority, used only when dynamic routing fails.
- **RIPng:** version of RIP adapted to IPv6, suited to small networks, configured per process and enabled per interface.
- **OSPFv3:** version of OSPF that supports IPv4 and IPv6, organised in areas, suited to larger networks.

Routing in IPv6 (2/2)

- Both protocols allow static routes to be injected and routes learned from other routing protocols to be redistributed.
- These protocols were designed for conventional networks; low-power, lossy networks require their own protocols, covered in the next lecture.

DHCPv6 and DNS in IPv6 (1/2)

- DHCPv6 allows centralised assignment of addresses and configuration parameters, including prefix delegation to downstream routers.
- When the server is not on the same link as the clients, a relay agent must be configured, which forwards requests to the server.
- Server configuration involves defining an address pool, the domain, the name servers and the lease times.
- In DNS, names now resolve to AAAA records, which hold 128-bit addresses, instead of the A records used in IPv4.

DHCPv6 and DNS in IPv6 (2/2)

- The same name may have both A and AAAA records, which lets IPv4 and IPv6 clients reach the same service during the transition phase.
- Reverse resolution uses the `ip6.arpa` zone, with the address written as hexadecimal digits in reverse order.

Lecture Summary (1/2)

- The 32-bit IPv4 address space is insufficient for the current scale of the Internet, and the Internet of Things makes the problem decisively worse.
- NAT allowed exhaustion to be postponed, but at the cost of breaking the end-to-end model, complicating protocols that carry addresses at the application layer and making it harder to reach internal devices from outside.
- IPv6 solves the problem structurally, with 128-bit addresses, a simplified header and no need for address translation.
- Stateless autoconfiguration is particularly valuable in IoT, because it lets devices obtain a valid address without manual intervention.

Lecture Summary (2/2)

- The Neighbor Discovery Protocol replaces ARP, using ICMPv6 messages and solicited-node multicast addresses, with efficiency gains over broadcast.
- The next lecture covers the adaptation of this protocol to networks of constrained devices, through 6LoWPAN.

Thank you.

Questions and comments